
10 — UI/UX Design System & GUI Polish Engine

This document catalogs the UI/UX design patterns, interactive micro-interactions, and visual components implemented to give V Foods a premium, state-of-the-art mobile feel.

1. Skeleton Shimmer Screens (Instant Perceived Performance)

- **Components:**
 - `MenuCardSkeleton.tsx` (`frontend/src/components/skeletons/MenuCardSkeleton.tsx`)
 - `OutletCardSkeleton.tsx` (`frontend/src/components/skeletons/OutletCardSkeleton.tsx`)
 - **What It Does:**
 - Displays pulsing gray placeholder blocks that match the exact shape of menu items (veg badge, title, price, image thumbnail, add button) while remote data loads.
 - **Why It Was Added:**
 - Eliminates layout shifting (Cumulative Layout Shift) when content loads.
 - Replaces clunky circular spinners with smooth, native-app-like loading states.
 - Makes the app feel 3x faster on slow hostel Wi-Fi and crowded mobile networks.
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2. Step-by-Step Live Order Tracker

- **Component:** `LiveOrderTracker.tsx` (`frontend/src/components/LiveOrderTracker.tsx`)
 - **What It Does:**
 - Renders an interactive 4-step progress timeline:
 1. **Order Placed:** Blue glow with clock icon.
 2. **Cooking & Prep:** Amber pulse with chef hat icon.
 3. **Ready at Counter:** High-contrast emerald glow with audio chime + vibration buzzer.
 4. **Collected:** Clean checkmark state.
 - Pinned automatically to the top of the student home screen whenever an order is in progress.
 - Includes a one-tap **"Pass QR"** button to present the scannable counter token.
 - **Why It Was Added:**
 - Eliminates student counter anxiety: students know whether food is still on the stove or ready on the counter, preventing physical crowding around the canteen pickup window.
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3. Spotlight Quick Search (`⌘K` / `Ctrl+K`)

- **Component:** `SpotlightSearch.tsx` (`frontend/src/components/SpotlightSearch.tsx`)
 - **What It Does:**
 - A full-screen dialog triggered via:
 - Desktop hotkey: `⌘K` on Mac or `Ctrl+K` on Windows/Linux.
 - Mobile: Tapping the "⌘K All Canteens" badge in the search bar.
 - Instant fuzzy search across **all 13 campus outlets simultaneously**.
 - Shows:
 - Dish Name & Price.
 - Veg / Non-Veg badge.
 - Which canteen serves it (Gazebo, AB3, North Square, etc.).
 - One-tap **"Add"** button with tactile haptic feedback.
 - Quick toggle: **"Veg Only"** filter.
 - **Why It Was Added:**
 - Solves the primary campus food question: *"Where can I get Cold Coffee or Samosa right now without checking 13 menus one by one?"*
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4. Mobile Bottom Sheet Drawer

- **Component:** `ItemDetailBottomSheet.tsx` (`frontend/src/components/ItemDetailBottomSheet.tsx`)
 - **What It Does:**
 - A smooth, iOS-style swipeable drawer powered by Framer Motion spring physics.
 - Slides up from the bottom of the screen with a swipe-down drag handle.
 - Displays:
 - High-resolution dish photography.
 - Veg/Non-Veg indicator and meal category tag.
 - Prep time estimate (e.g. *5–8 mins prep*), calorie count (*~240 kcal*), and freshness badge.
 - Stepper quantity selector (`- 1 +`).
 - Full-width "Add to Tray" button with live price calculation.
 - **Why It Was Added:**
 - 95%+ of students use mobile phones. Standard boxy centered modal popups feel like legacy desktop websites; bottom drawers provide a natural thumb-friendly native app experience.
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5. Web Haptics Engine (`useWebHaptics`)

- **Hook:** `useWebHaptics.ts` (`frontend/src/hooks/useWebHaptics.ts`)

- **What It Does:**
 - Utilizes the native HTML5 `navigator.vibrate()` API with graceful fallbacks.
 - Provides distinct tactile patterns:
 - **tap (10ms)**: Light click when adding a dish to the cart or tapping quantity steppers.
 - **success (20ms-40ms-20ms)**: Double-beat affirmation when an order is placed.
 - **ready (50ms-50ms-100ms)**: Attention-grabbing pulse when the kitchen marks the token ready for pickup.
 - **error (80ms-40ms-80ms)**: Warning buzz for insufficient wallet balance or stock-out.
 - **Why It Was Added:**
 - Gives the browser-based PWA the physical tactile responsiveness of native iOS and Android apps without requiring a 100MB app-store download.
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6. KDS Ticket Aging Badges (Kitchen Staff Dashboard)

- **Implementation:** `getAgingBadge()` in `ShopStaffDashboard.tsx`.
 - **What It Does:**
 - Automatically calculates elapsed prep time and changes ticket border & badge colors:
 - **Green (#ECDF5)**: < 5 minutes in queue (Fresh ticket).
 - **Amber (#FFFBEB)**: 5–10 minutes in queue (Standard cooking window).
 - **Flashing Red (#FEF2F2)**: > 10 minutes in queue (Delayed order alert).
 - **Why It Was Added:**
 - Canteen kitchens during break rushes are loud and hectic. Kitchen staff can spot delayed tickets from 10 feet away and prioritize older tickets without touching the screen.
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7. Night Canteen OLED Dark Theme

- **Design:** True deep black backgrounds (`#090D16`), subtle slate borders (`#1E293B`), and glowing emerald/blue accents.
 - **Why It Was Added:**
 - Campus night canteens run from **10:00 PM to 1:30 AM**. Students ordering from dark dorm rooms get an eye-friendly, battery-saving dark interface.
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